A TWO-PERIODIC COMPLEX FORMED FROM TWO INJECTIVE ENDOMORPHISMS WITH COMMUTING IMAGES OVER A ONE DIMENSIONAL NOETHERIAN LOCAL RING

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-chow/preamble.tex is excerpted verbatim as two spans, lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 02PX (sources/stacks-chow/pinned-original-source.tex line 17371).

Allowed standard prerequisites (editorial).

- Support of a finite module over a Noetherian local ring and finiteness of length at a minimal prime of the support. Algebra, Lemma labelled algebra-lemma-support-point (Stacks tag 00L5), invoked at proof line 17385. Supplies that M_q has finite length at a minimal prime q of the support and that a module whose support is the closed point has finite length; the exactness computation is inside the counted span.
- Two-periodic complexes and their exactness. Same chapter, chow.tex, the definition of a (2, 1)-periodic complex and of exactness used in the statement. Fixes the notion whose two kernel conditions are verified in the counted proof.

Source reference. Original lemma 00L5: sources/stacks-chow/tags.txt, source lines 775–775. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 17385 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *Let A be a Noetherian local ring. Let M be a finite A-module of dimension 1. Assume $\varphi, \psi : M \to M$ are two injective A-module maps, and assume $\varphi(\psi(M)) = \psi(\varphi(M))$, for example if φ and ψ commute. Then $\text{length}_A(M/\varphi\psi M) < \infty$ and $(M/\varphi\psi M, \varphi, \psi)$ is an exact $(2,1)$-periodic complex.*

Proof. Let $\mathfrak{q}$ be a minimal prime of the support of M. Then $M_\mathfrak{q}$ is a finite length $A_\mathfrak{q}$-module, see Algebra, Lemma 00L5. Hence both φ and ψ induce isomorphisms $M_\mathfrak{q} \to M_\mathfrak{q}$. Thus the support of $M/\varphi\psi M$ is $\{\mathfrak{m}_A\}$ and hence it has finite length (see lemma cited above). Finally, the kernel of φ on $M/\varphi\psi M$ is clearly $\psi M/\varphi\psi M$,

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and hence the kernel of φ is the image of ψ on $M/\varphi\psi M$. Similarly the other way since $M/\varphi\psi M = M/\psi\varphi M$ by assumption. $\square$